

Anil Seth



Choosing Between Replacing and Renewing a Desktop

“Renew the desktop using the advice of Linus Torvalds—boot from a Solid-State Disk.”

My desktop seemed slow. It is four years old and has faced a fair amount of criticism from my family. I toyed with the idea of buying a new one. Moore's Law is supposed to still hold good, but I found that the extra performance is via more cores. Each core is not much faster. Hence, the new desktop may not be any faster, at least, for booting and signing in. And what would I do with the older one? I had installed Fedora when I bought the system. Since then, I have been upgrading it. Could something have gone wrong? I freed a partition and installed a fresh copy of Fedora 17 on it. The new installation was much snappier. The boot time, as shown by *systemd-analyze*, came down from 72 to 27 seconds!

However, the comparison wasn't fair. The original installation had a lot of services installed on it. Removing the additional services reduced the boot time to 45 seconds. The kernel and the *initramfs* timings were about the same. So what was the difference? There may be some issues related to repeated upgrades of the distribution, which are not the supported methods. While searching for differences, I did find an explanation for why the Ethernet interface was still named *eth0* and not *em0* as per the default Fedora installation. During the upgrades, an additional package called *biosdevname* was introduced, yet it was ignored. No package required it; so, its use was optional. Besides, the device name difference was not relevant as far as performance was concerned.

Disk fragmentation can have an impact on performance. While de-fragmentation is not supposed to be a necessity on Linux, various upgrades and experimentations may have seriously fragmented the root partition. Backing up the root partition, reformatting and restoring it reduced the boot time to 34 seconds.

An SSD as root

There was a parallel approach. I had been fascinated by articles about dropping Solid-State Disk prices. Unfortunately, the local vendors had no idea, or quoted a price that was still high. Recently, I found an online store offering 60 GB at prices not too far above the US prices. In a remarkable coincidence, the SSD disk was delivered to me the same day as I read the, “Get thee behind me, Satan!” comment by Linus Torvalds on *slashdot.org*.

The SSD disk behaves just like a SATA disk, so installing and using it as an additional disk was not a problem. However, it would not boot from the SSD disk. It was obviously a hardware limitation of my old desktop. The workaround was to create a boot partition on a conventional disk, and have the root partition on SSD. The result was a remarkable improvement in the

responsiveness of the system. *Systemd-analyze* showed the new boot time as 14 seconds. This figure is even more remarkable when you notice that the kernel and *initramfs* timing is still the same as original, which is about 7 seconds. So, the user-space boot time has come down from 38 to 27 seconds after de-fragmentation, and to 7 seconds with SSD!

Bootchart

A very useful utility for identifying boot performance is the *bootchart* package. It collects the boot data, and prepares a beautiful, detailed chart—*bootchart.png* in the */var/log* directory. Unlike *systemd-analyze*, this includes the start of the display manager. This is the time that matters to the user. With the SSD, *bootchart* gives the time as 26 seconds when the login screen is ready. The corresponding timing for a SATA root disk is 56 seconds originally, and 41 seconds after de-fragmentation. The difference between the two types of disks can be seen from the following table:

Cumulative time	SSD root	SATA root	SATA root restored
CPU time	24 sec	29 sec	28 sec
IO time	44 sec	343 sec	177 sec
Elapsed time	26 sec	56 sec	41 sec

The conclusion is obvious. I/O time is the most significant variable in reducing boot time. The benefit of no rotational delay and minimal uniform access time in a SSD disk gives a new life to an old system.

A 60 GB SSD is a cost-effective option in a desktop, as it can be used for root and home. Multimedia and other large files can be on the existing SATA or IDE hard disk. This investment is likely to result in better performance than replacing the desktop—unless, of course, your brand-new desktop relies exclusively on SSD.

My next project is to refresh my netbook. That may happen soon, as more options for 120 GB SSDs are becoming available in India, and local prices are likely to become comparable to US prices. 

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